

PAP521S - PROGRAMMING IN PRACTICE

Project A: Individual Contribution Record

Student Name: Marthin Siteketa

Student Number: 225030969

Course Code: PAP521S

Project Title: Municipal Financial Management System (MFMS)

Group Role: Documentation, Testing and Group Technical Report Author

1. ASSIGNED RESPONSIBILITY

My primary responsibility on Project A was leading system documentation and managing overall testing. Specifically, I authored the official 5-page Group Technical Report, compiled software testing results, and documented user workflow requirements based on the system features developed by our team.

2. FUNCTIONS AND MODULES DEVELOPED

My main contribution focused on system documentation, verification, and technical writing:

- **Short Technical Report:** Wrote and formatted the complete 5-page Group Technical Report covering all required sections: Introduction, Problem Description, System Objectives, System Features, Program Design, Challenges Encountered, Solutions Implemented, and Conclusion.
- **Documentation and Guides:** Drafted content for the project documentation, detailing the system overview, compilation instructions using GCC, menu navigation, and module summaries.
- **System Testing and Validation Scenarios:** Designed manual test scenarios to verify menu options and data validation rules, such as preventing negative budgets or salaries and testing boundary choices.

3. GITHUB CONTRIBUTION

- **Documentation Commits:** Pushed commits containing the project report, documentation updates, and technical notes directly into the group repository created by our team lead.
- **Collaborative Updates:** Pulled code updates from group members to verify that the integrated software matched the workflow described in our written report.

4. TESTING PERFORMED

I carried out systematic manual testing on the built program to ensure the system functioned according to assignment guidelines and to document the results in our report:

- **Menu Navigation and Input Validation:** Tested main menu input handling using valid numbers (1 to 6) as well as invalid inputs (letters, negative numbers, and numbers out of range). Confirmed that negative numbers entered for basic salaries, allowances, or budgets were properly rejected.
- **Module Functionality and Calculations:** Checked salary calculations and departmental remaining budget calculations (Allocated Budget minus Expenditure) across sample data inputs. Tested search functions for employees, suppliers, and assets to confirm matching logic and error warnings when items were not found.